

CLASS PARTICIPATION COVERSHEET

Unit of Study: **MIT653 Research Methods**
Assignment Title: **Class Participation – Week 03**
Semester and Year: **Trimester 2, 2026**
Lecturer Name: **Dr. Sudath Heiyanthuduwege**

DECLARATION

The work contained in this assignment is my own work and has not been copied from other sources or been previously submitted, unless otherwise specifically acknowledged.

I understand that failure to comply can lead to severe penalties.

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Compliance: _____

Date Submitted: ____dd____mm____yy

Signed: _____ Date: ____dd____mm____yy

EXAMINER ONLY

Due Date: ____dd____mm____yy

MARK: _____

FINAL MARK: _____

Class Participation – Week 3: Literature Review

Section A:

1. Role of a comprehensive literature review in the research process

A literature review plays a foundational role in research. It helps establish background knowledge on the topic, summarises and synthesises what is already known, identifies gaps and inconsistencies where further research is needed, and helps build the theoretical framework for a new study. As covered in the Week 3 lecture, it can be thought of as a pyramid — starting broad (general context and background) and narrowing down to specific theories, methodologies, and finally the researcher's own research question. It is not simply a summary of everything read, but a critical synthesis that connects existing knowledge to the researcher's own contribution.

2. Key steps of a systematic literature review

A systematic literature review follows three broad phases: **Plan Review** (defining the scope, developing a search strategy and inclusion/exclusion criteria, managing resources and databases), **Conduct Review** (searching the literature, analysing and evaluating it, synthesising findings, and identifying gaps and trends), and **Document Review** (structuring the review, writing it up, and citing sources properly). This structured approach ensures the review is rigorous, reproducible, and unbiased.

3. Using a literature review to identify gaps and suggest future research

By critically comparing multiple studies, a literature review reveals where findings conflict, where methodologies are weak or outdated, and where certain aspects of a topic remain unexplored. In my own review of "Use of AI in Cybersecurity Threat Detection," comparing three papers showed recurring gaps — such as poor model explainability, reliance on outdated datasets, and vulnerability to adversarial attacks — which directly point toward meaningful directions for future research.

4. Role of the research question in conducting a literature review

Yes, the research question is central to the literature review process. It guides what to search for, which keywords and databases to use, and which studies are relevant enough to include. As shown in the lecture's pyramid model, the review process narrows from broad background information down to the specific research question, and the final stage of the review explicitly connects existing literature back to that question — explaining how the new study will contribute to the field.

Section B:

For this task, I identified and reviewed three papers relevant to my research topic:

1. **Buczak & Guven (2016)** – *A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection* — a foundational survey establishing core ML methods used in intrusion detection.
2. **Rehman et al. (2025)** – *A Systematic Literature Study of Machine Learning Techniques Based Intrusion Detection* — a recent, comprehensive PRISMA-based review of 104 papers covering datasets, models, and future directions.
3. **Adversarial Machine Learning in Network Intrusion Detection Domain: A Systematic Review (arXiv)** — highlights how AI-based detection systems can themselves be deceived by adversarial attacks.

Across these papers, common themes emerged: a clear shift from traditional signature-based security to AI/ML-based detection, heavy reliance on (often outdated) benchmark datasets, and a persistent trade-off between detection accuracy and computational/interpretability cost. Key research gaps identified include the lack of real-time scalable models, limited explainability of deep learning-based systems, and weak robustness against adversarial manipulation — all of which help define potential directions for my own research.

The full analysis has been documented in LiteratureReview.md on my GitHub repository: [MIT653-Research-Project](#).